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BITS Pilani
Pilani Campus



Autotronics

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AEZG511/AELZG511

Lecture No.1-Introduction

Electric and Hybrid Vehicles



- Faculty: Dr. Madhuri Bayya, Prof.Rakesh Dash
- TAs
- Lab: Prof.Rakesh Dash
- Class Material:
 - To be posted in Teams
 - Drive link for textbooks:
 - https://drive.google.com/drive/folders/1Z4iCdV2TYk6GO_GjD1zGPTHwjp3Miclt?usp=drive_link
- Quiz/Assignment
 - All will be done through e-learn portal: <https://elearn.bits-pilani.ac.in/>

Evaluation Components

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weightage	Duration
EC – 1*	Situational learning Assignment/QUIZ	Online	10%	
	Lab	Online	20%	
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours
EC - 3	Comprehensive Exam	Open Book	40%	2 ½ Hours

Vehicle

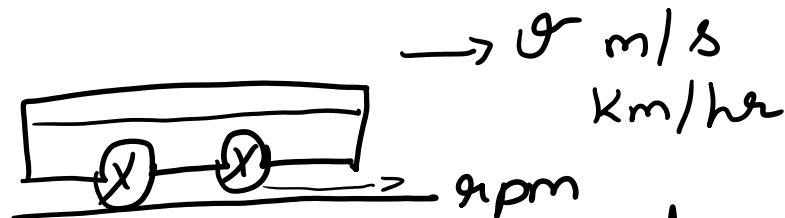
↳ Measurable quantity

↳ fuel, speed, air, temperature,
pressure, rpm, voltage, current,
% throttle

↳ time dependent

↳ Mechanical, electrical signal

↳ Input/output



$$v = r \omega$$

↓
radius

↳ angular velocity (ω)
rad/s

AUTOTRONICS COURSE

Measured quantity

Types of signals

How?

Sensor / Transducers

Types

Electrical signal
(mV or μ V) → analog
(voltage) → digital

Amplification / attenuation
analog

ADC → analog to digital converter

Signal Conditioning (SC)

μ C

ECU

open
↑
Control
↑
Closed

DAC (SC)

Actuators
Types

OP

↳ Modelling

↳ Mathematical

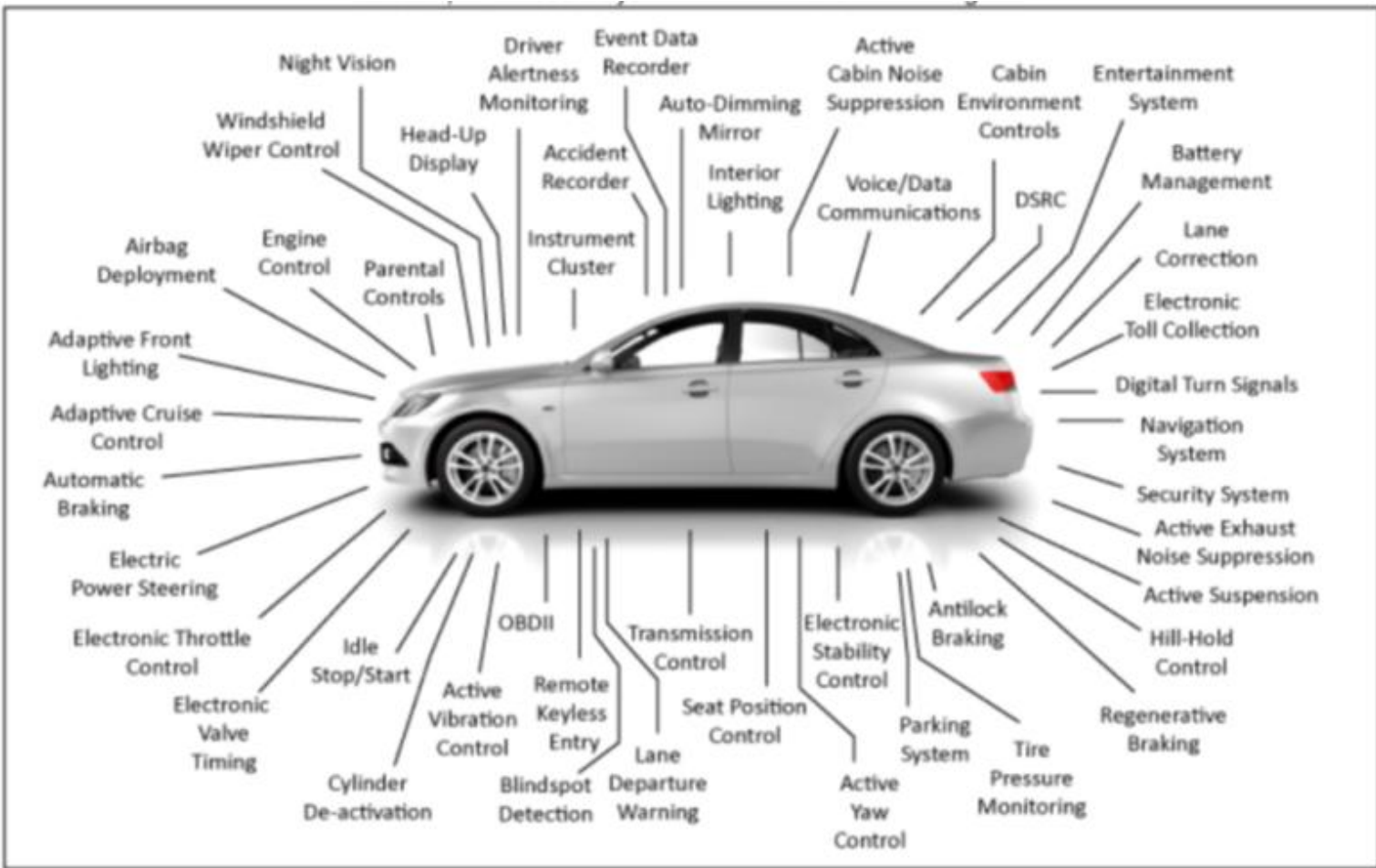
↳ Examples

Signal $\rightarrow$ Electrical $\rightarrow$ amplified
 of P
 0-200km/h (0-5) V.

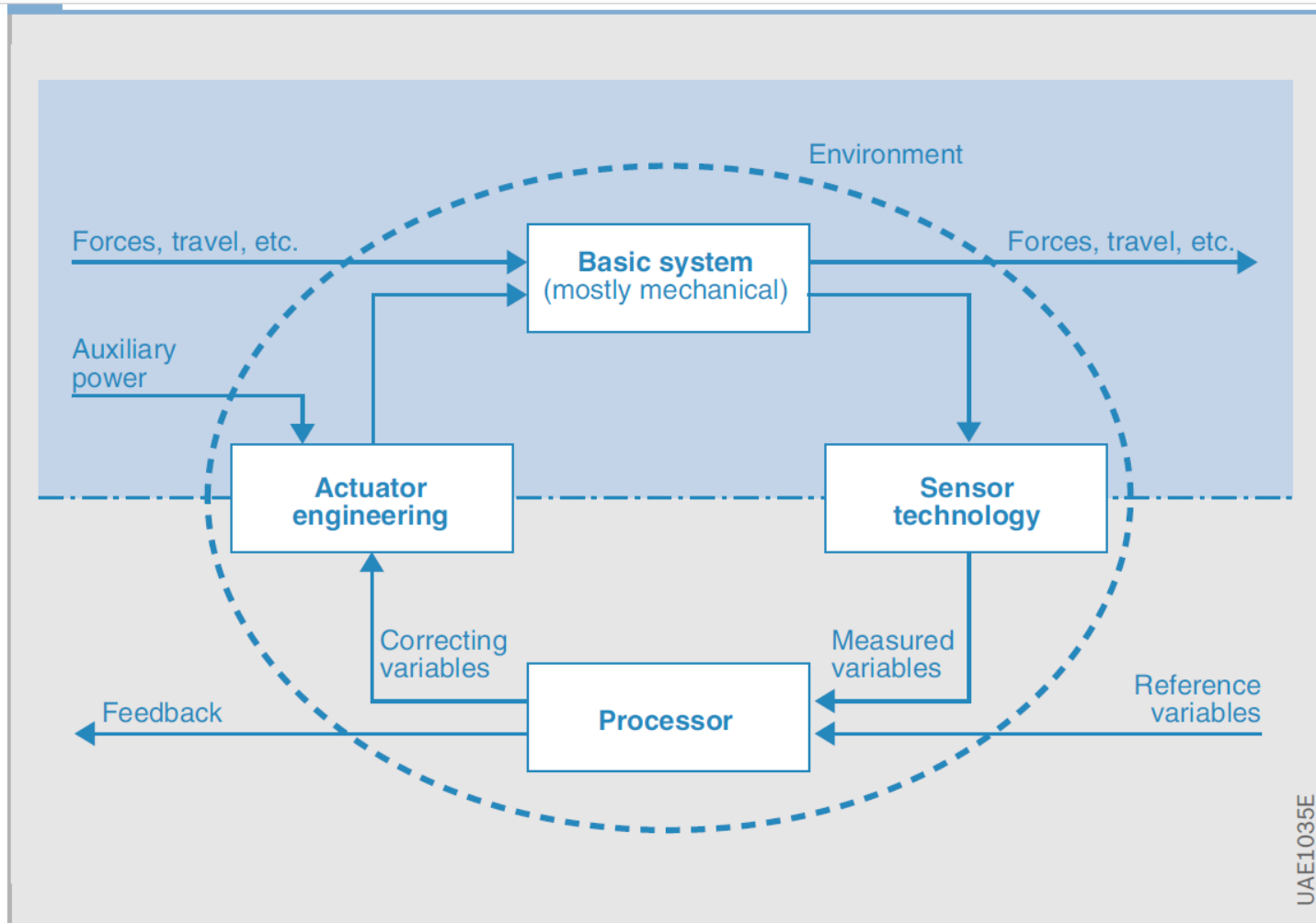
Aspects of sensors & SC

1. Accuracy
2. Resolution
3. Calibration
4. Error.
5. Range/ Span
6. Precision.

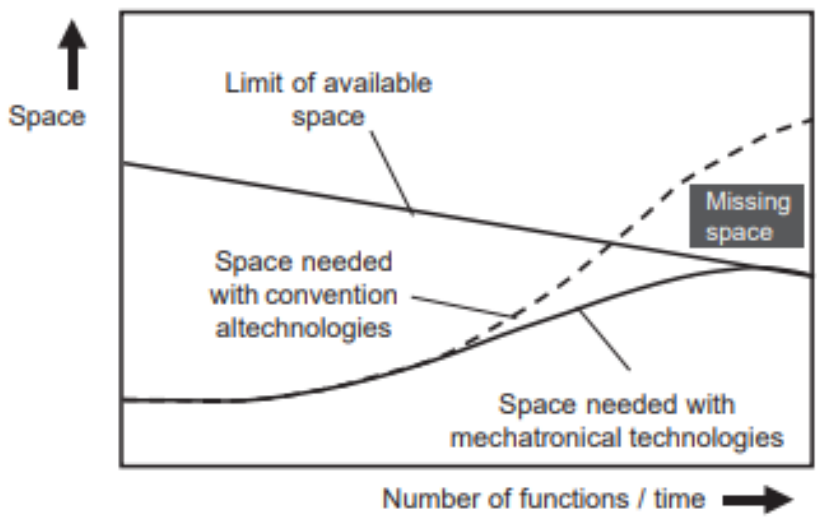
Static &
 dynamic
 characteristics.



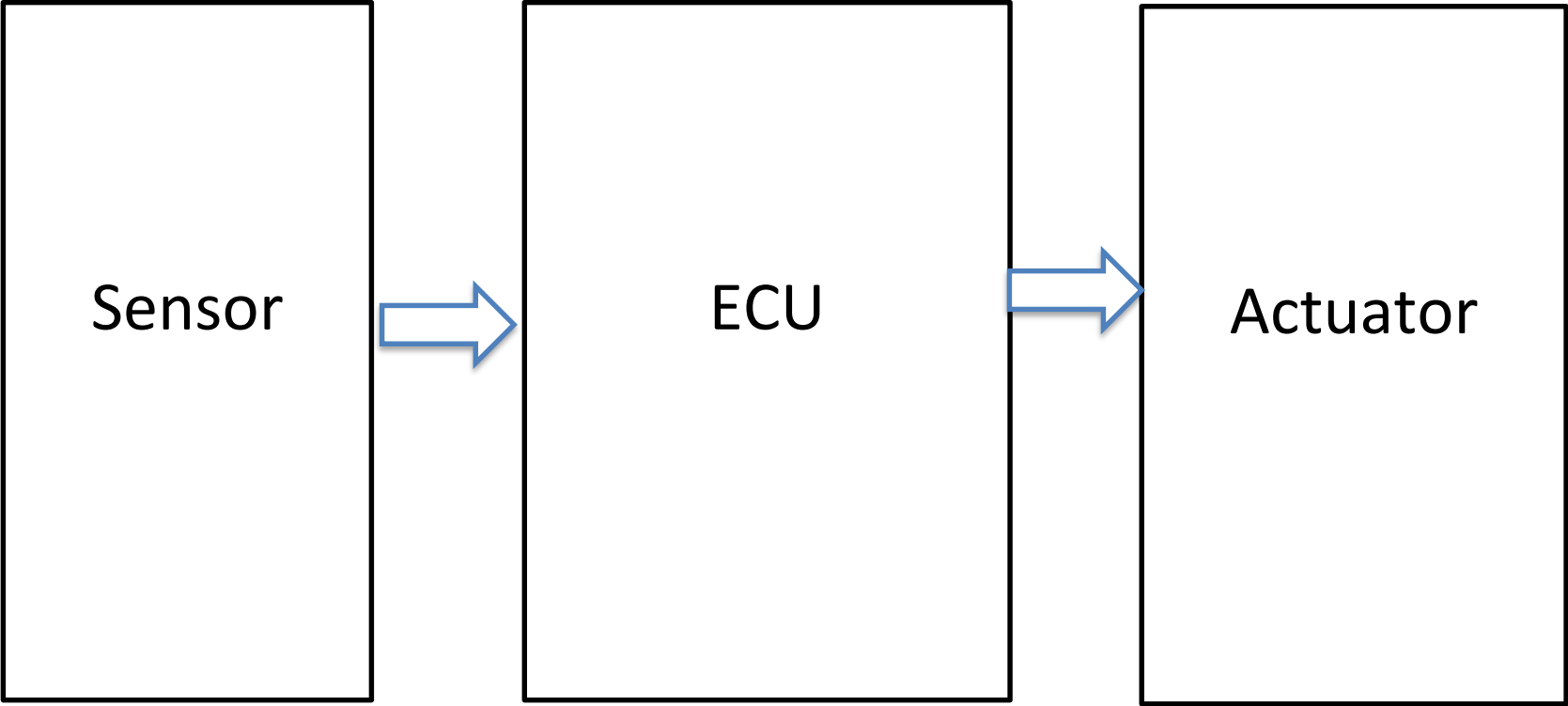
What is it?



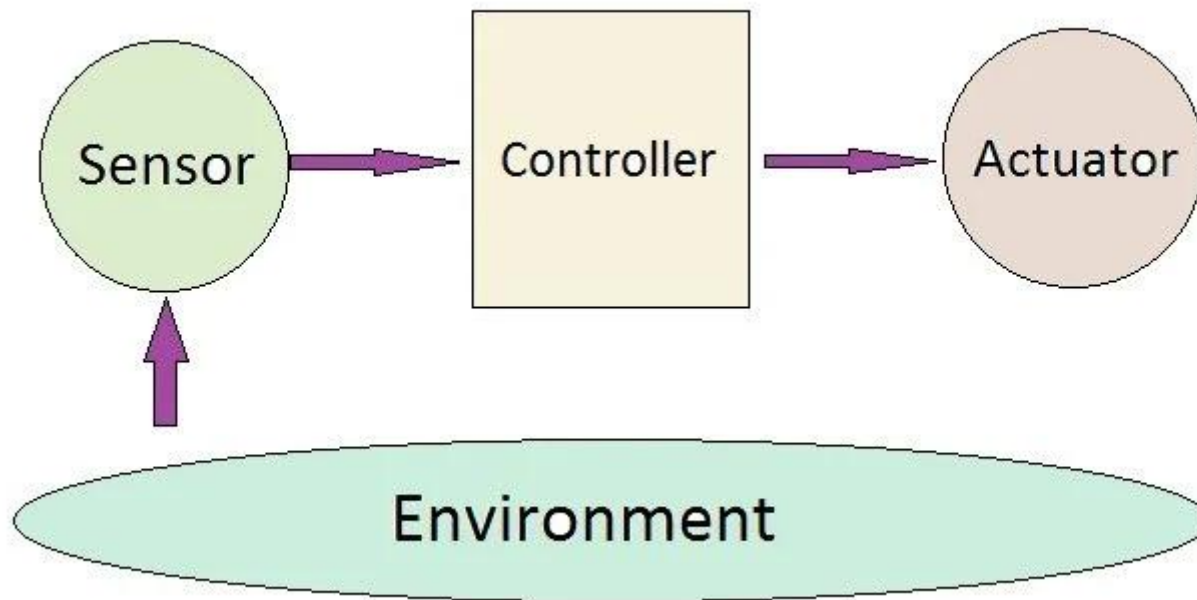
Space in vehicle



Vehicle automation and control



Control System



Electronic Control Units (ECUs)



- ❑ The ECU, which refer to a single module or a collection of modules, monitor and control different functions in the vehicle – they may be standard from the manufacturer or reprogrammable
- ❑ Tuning features of the ECU allow the user to set the operating mode of the engine – economy, performance etc.



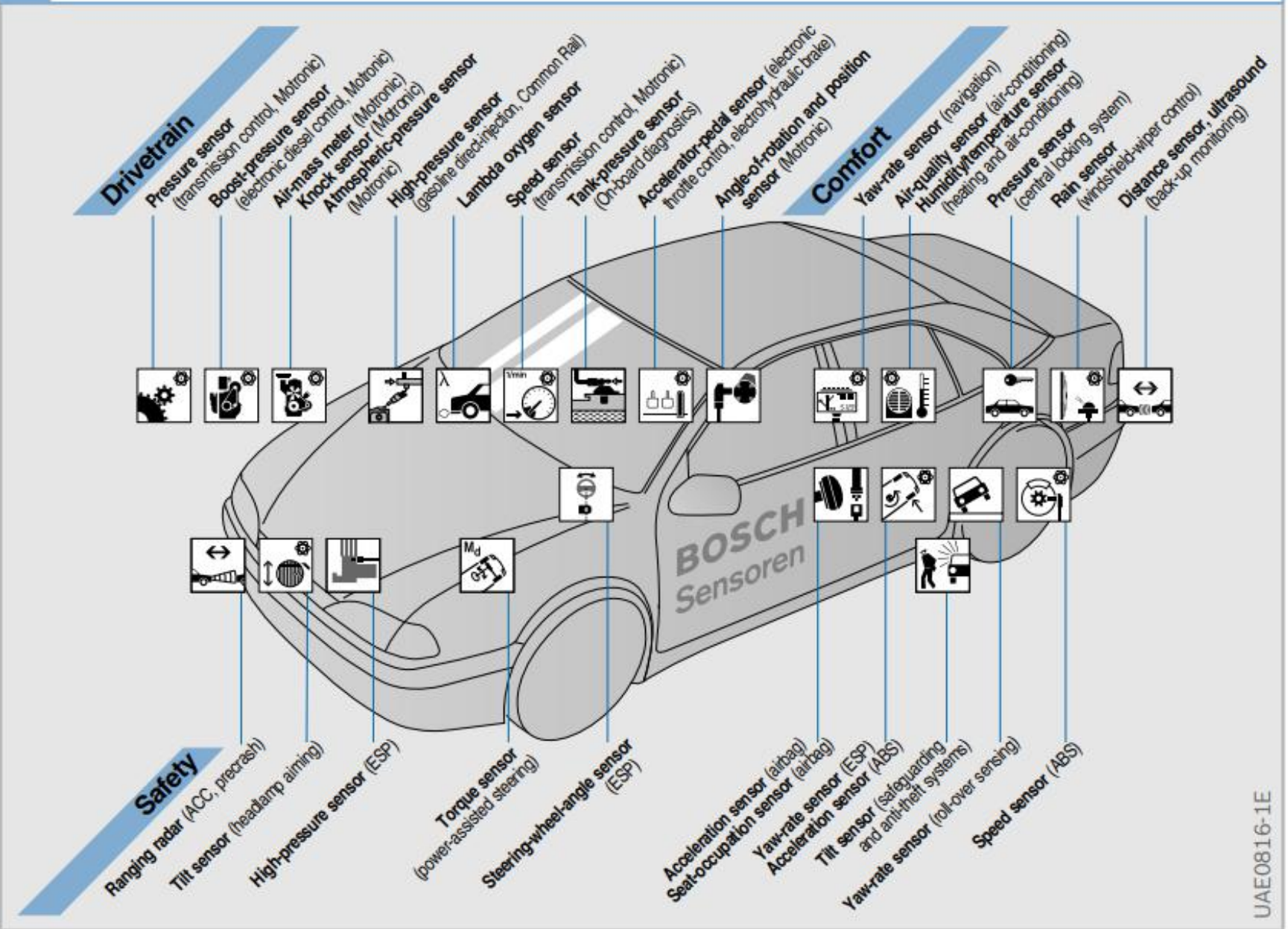
↳ ICE & EV subsystems



The various ECUs in a vehicle

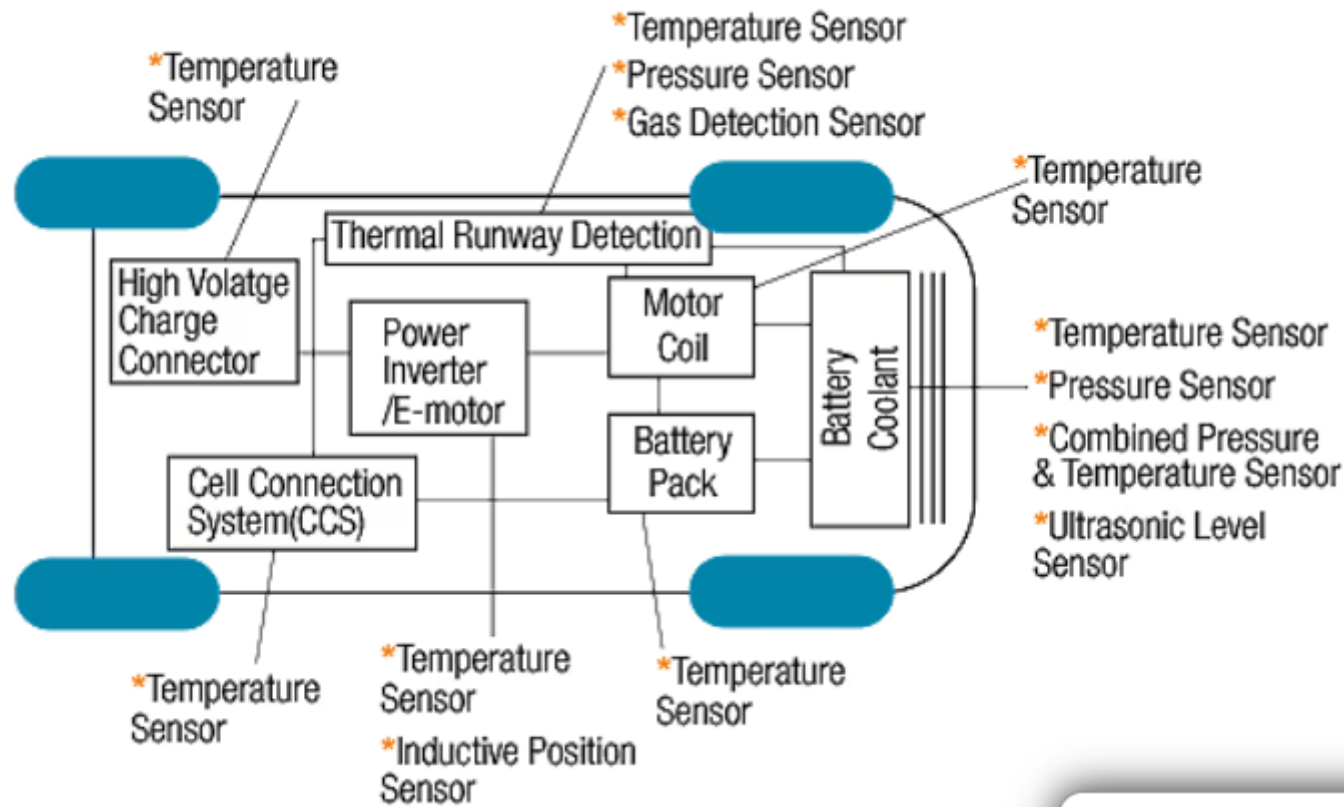
- ✓ Engine Control Module (ECM): Controls the actuators of the engine to set ignition timing, A/F ratio, idle speed etc.
- ✓ Vehicle Control Module (VCM): Stability control, traction control, yaw control etc.
- ✓ Transmission Control Module (TCM)
- ✓ Powertrain control module (PCM) – Combination of ECM & TCM
- ✓ Electronic Brake Control Module (EBCM): Controls the Antilock Braking System (ABS)
- ✓ Body Control Module (BCM): Controls body features – power windows, power seats...

1 Automotive systems and their sensors

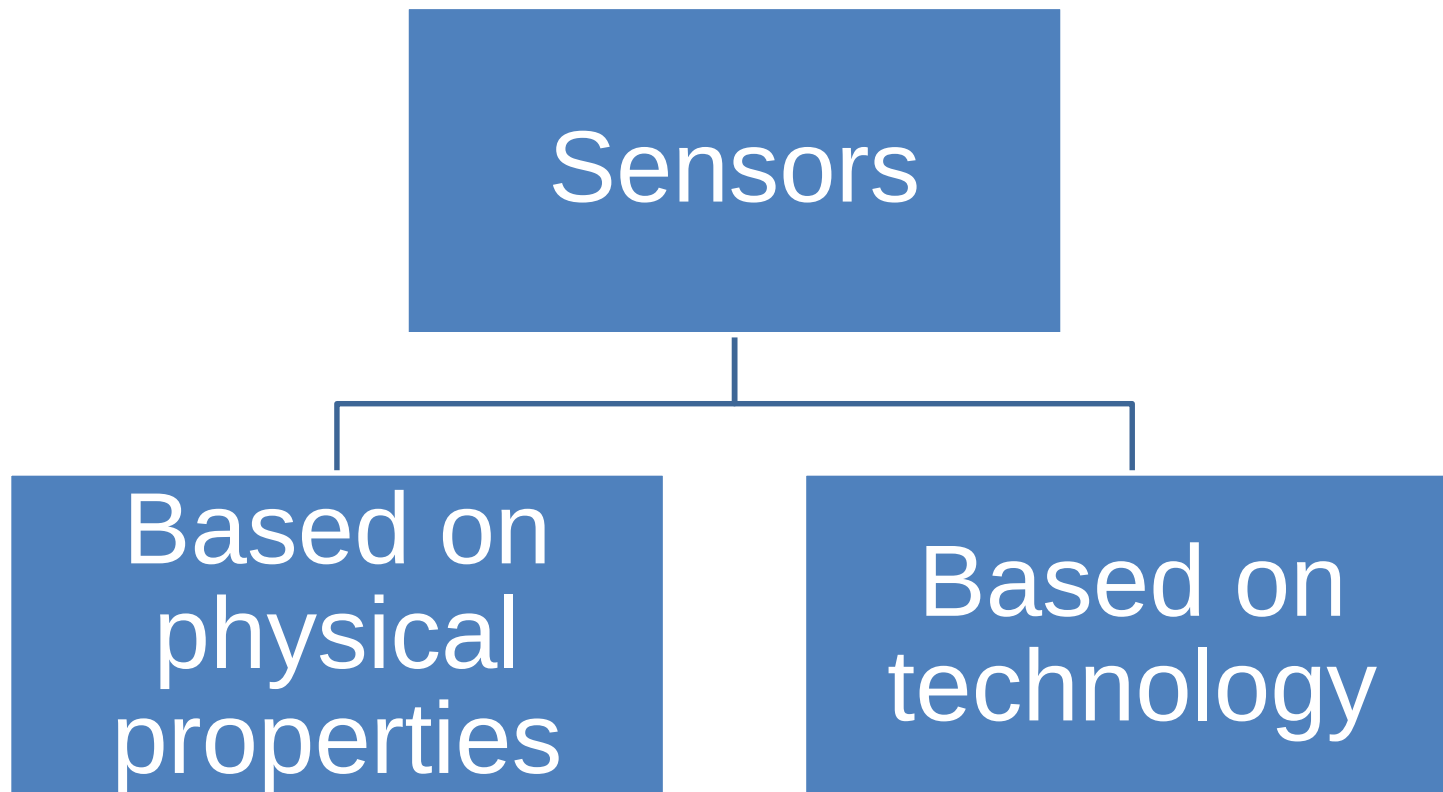


UAE0816-1E

EV/HEV



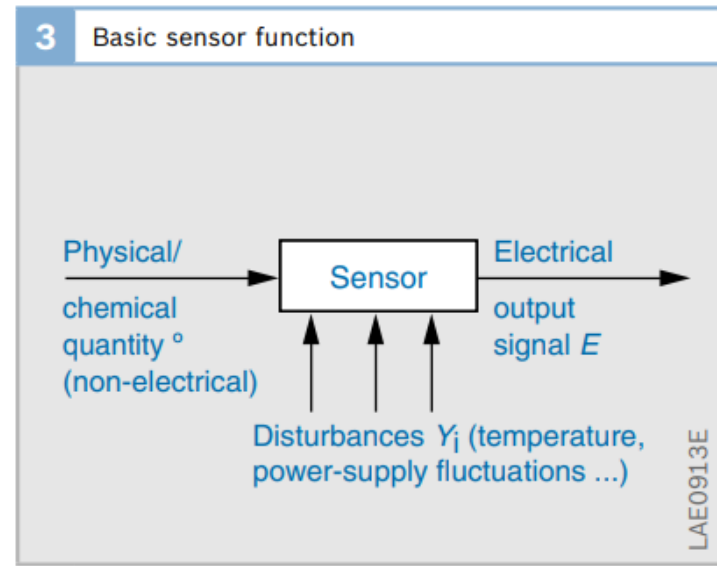
Classification



Specifications

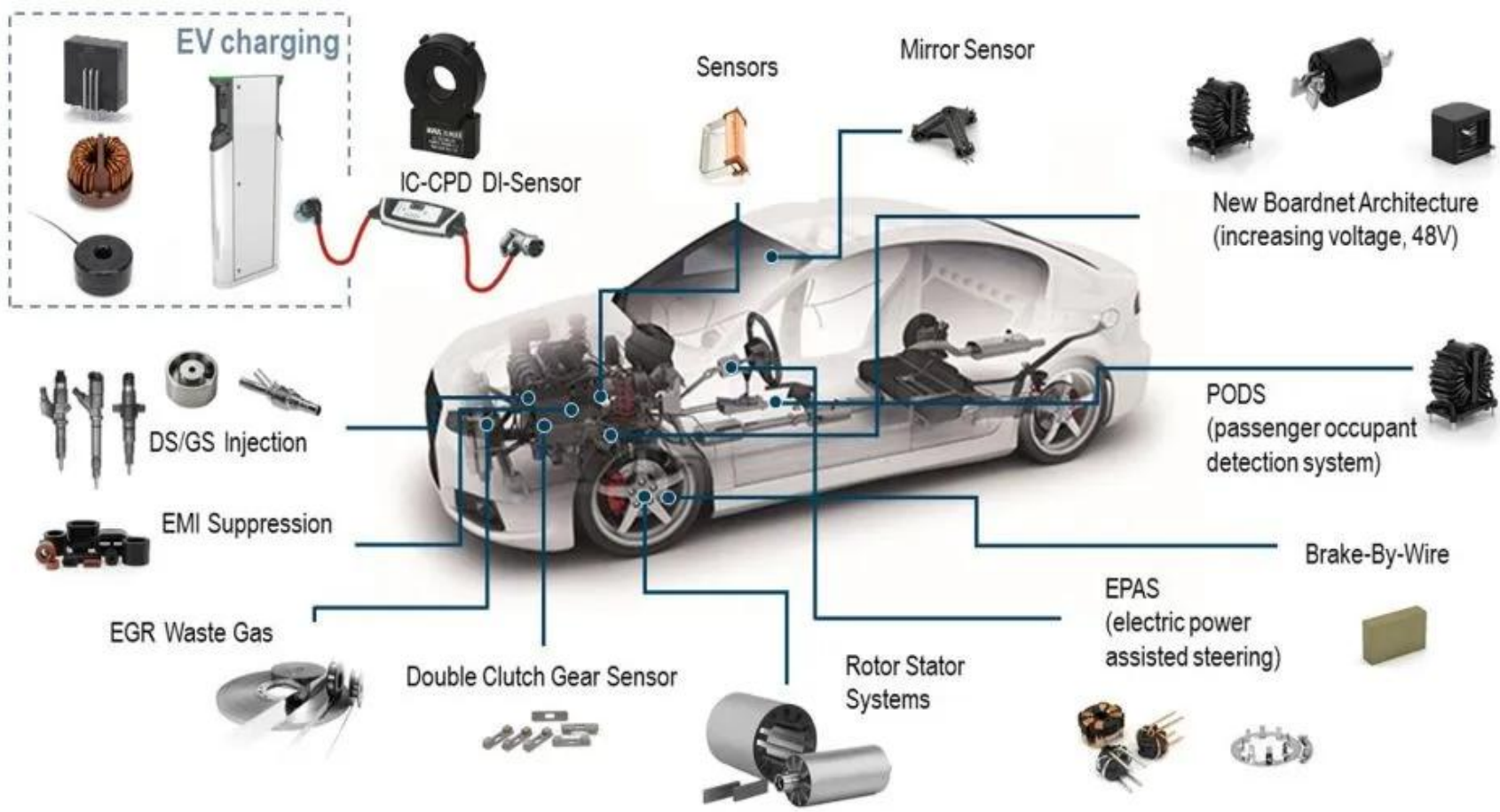


- Accuracy and resolution
- Sensitivity and Range
- Environmental considerations
- Sensor tolerances
- Fault Diagnostics

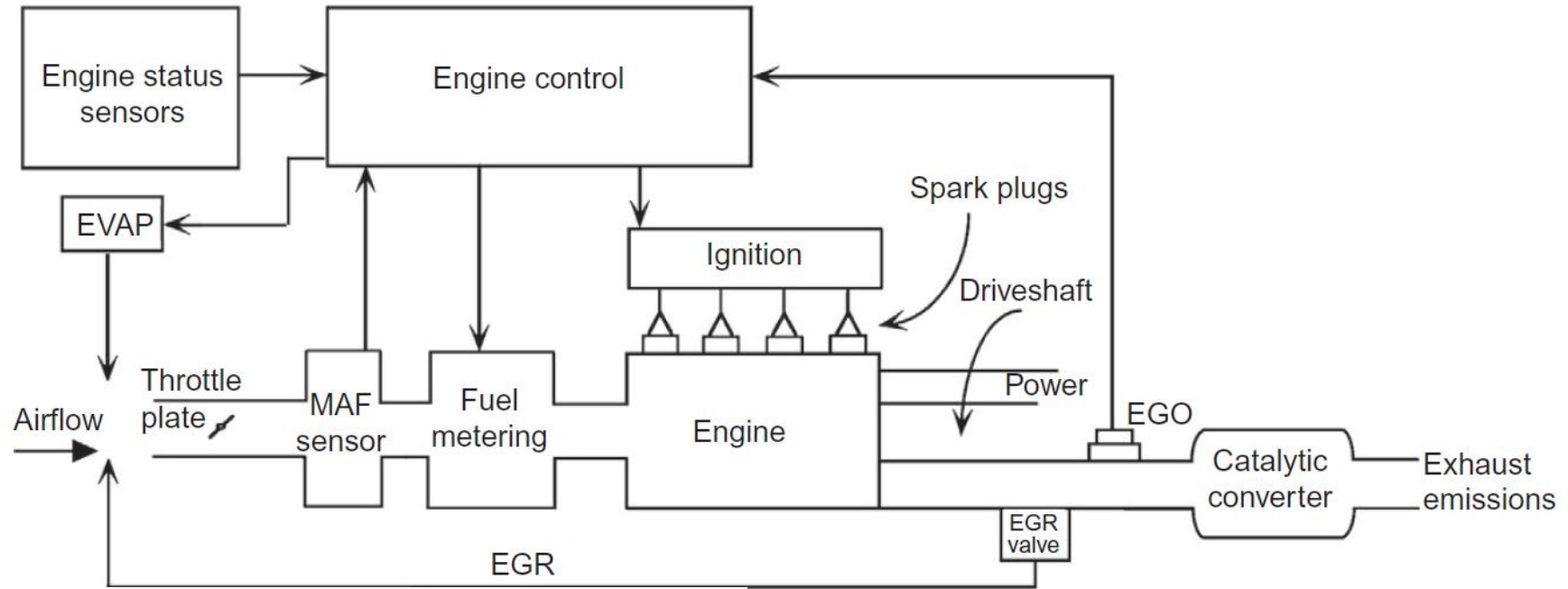


Building blocks - Signals

Analogue	Digital
<u>Advantages</u>	<u>Advantages</u>
Produces a more 'faithful' reproduction of the physical quantity.	Can be very immune to noise
Usually simple	Signal can be transmitted over long distances.
<u>Disadvantages</u>	<u>Disadvantages</u>
Noise and distortion problems	Output subject to quantity errors from sampling
	Can be complex

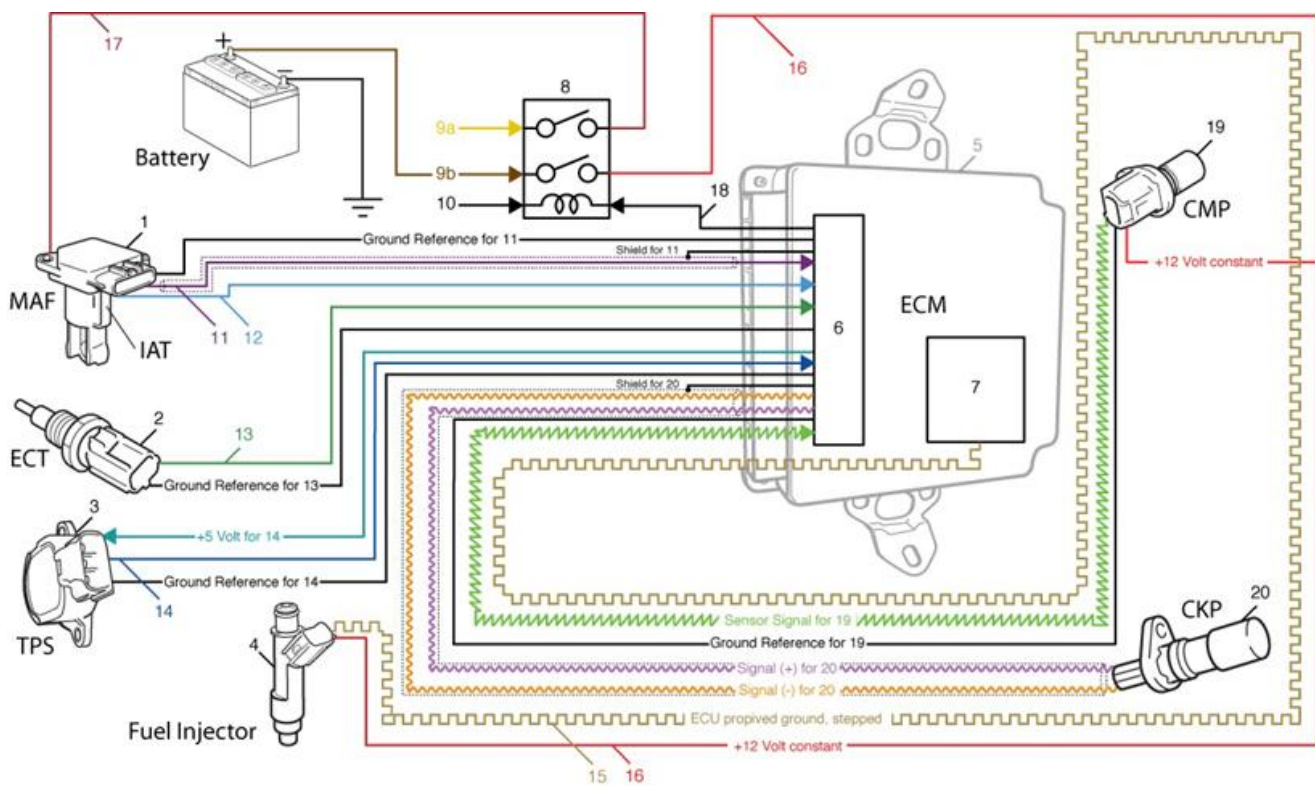


Building blocks - Systems



1. Throttle position sensor (TPS)
2. Mass airflow rate (MAF)
3. Engine temperature (coolant temperature) (CT)
4. Engine speed (RPM) and angular position
5. Exhaust gas recirculation (EGR) valve position
6. Exhaust gas oxygen (EGO) concentration

Building blocks - Systems



1. Mass Air Flow w/ IAT Sensor (MAF)	8. Main or EFI relay	14. Throttle Position Sensor (TPS) sensor signal
2. Engine Coolant Temp (ECT) Sensor	9a. From "EFI or Main" fuse +5v	15. Drive Signals for fuel injector(s), ECU provides ground path
3. Throttle Position Sensor(s) (TPS)	9b. From "EFI or Main" SBF fuse +12v	16. Power Supply for fuel injector(s) and CMP sensor, +12v switched
4. Fuel Injector(s)	10. From Main relay	17. Power Supply MAF and IAT, +5v switched
5. Engine Control Module (ECM)	11. Mass Air Flow (MAF) sensor signal	18. Control signal for Main or EFI relay
6. Central Crossing Unit (CPU)	12. Intake Air Temp (IAT) sensor signal	19. Camshaft Position (CMP) sensor
7. Drive circuit for Fuel Injectors	13. Engine Coolant Temp (ECT) sensor signal	20. Crankshaft Position (CKP) sensor